

10- پیوست شماره 2: کدهای اجرای سناریو Apache Hadoop و اجرای Classifier**execute-with-RandomForest.py**

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#!/usr/bin/env python3
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import os
import sys
import time
import subprocess
import pandas as pd
import numpy as np
from pathlib import Path
from datetime import datetime
from sklearn.ensemble import RandomForestClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import accuracy_score
import warnings
warnings.filterwarnings('ignore')

HADOOP_HOME = "/usr/local/hadoop"
HDFS_INPUT_DIR = "/user/amos/experiments/input"
HDFS_OUTPUT_BASE = "/user/amos/experiments/output"
LOCAL_RESULTS_DIR = "./experiment_results"
LOCAL_DATASET_PATH = "./normalized_dataset.csv"

EXPERIMENTS = [
    (1, 1, 1, 100, "baseline_small_chunk"),
    (2, 1, 1, 200, "baseline_medium_chunk"),
    (3, 1, 1, 500, "baseline_large_chunk"),
    (4, 1, 1, 1000, "baseline_full_chunk"),
    (5, 1, 3, 1000, "more_reducers"),
    (6, 1, 5, 1000, "even_more_reducers"),
    (7, 3, 1, 1000, "more_mappers"),
    (8, 3, 3, 1000, "balanced_equal"),
    (9, 3, 5, 1000, "balanced_more_reducers"),
    (10, 5, 1, 1000, "even_more_mappers"),
    (11, 5, 3, 1000, "balanced_2"),
    (12, 5, 5, 1000, "balanced_full"),
]
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]

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CLASSIFIERS = {
    'Random Forest': RandomForestClassifier(n_estimators=100, random_state=42,
n_jobs=-1),
    'Decision Tree': DecisionTreeClassifier(random_state=42),
    'KNN': KNeighborsClassifier(n_neighbors=5, n_jobs=-1),
    'Naive Bayes': GaussianNB(),
    'SVM': SVC(kernel='rbf', random_state=42)
}

def create_directories():
    Path(LOCAL_RESULTS_DIR).mkdir(parents=True, exist_ok=True)
    for i in range(1, 13):
        Path(f"{LOCAL_RESULTS_DIR}/exp_{i:02d}").mkdir(parents=True,
exist_ok=True)

def check_hadoop():
    result = subprocess.run(["which", "hadoop"], capture_output=True, text=True)
    if result.returncode != 0:
        print("[ERROR] Hadoop not found in PATH")
        return False
    print(f"[OK] Hadoop found: {result.stdout.strip()}")
    return True

def check_hdfs():
    result = subprocess.run(["hdfs", "dfs", "-ls", "/"], capture_output=True,
text=True)
    if result.returncode != 0:
        print("[ERROR] HDFS not available")
        return False
    print("[OK] HDFS is available")
    return True

def setup_hdfs():
    subprocess.run(["hdfs", "dfs", "-mkdir", "-p", HDFS_INPUT_DIR],
capture_output=True)
    subprocess.run(["hdfs", "dfs", "-rm", "-r", "-f", HDFS_OUTPUT_BASE],
capture_output=True)
    print(f"[OK] HDFS directories prepared: {HDFS_INPUT_DIR}")
    return True

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def load_full_dataset():
    df = pd.read_csv(LOCAL_DATASET_PATH)

    if 'label' in df.columns:
        label_column = 'label'
    elif 'Label' in df.columns:
        label_column = 'Label'
    else:
        label_column = df.columns[-1]

    feature_columns = [col for col in df.columns if col != label_column]
    full_features = df[feature_columns].values
    full_labels = df[label_column].values

    return full_features, full_labels, feature_columns


def upload_csv_to_hdfs():
    hdfs_csv_path = f"{HDFS_INPUT_DIR}/data.csv"
    subprocess.run(["hdfs", "dfs", "-rm", "-f", hdfs_csv_path],
capture_output=True)
    result = subprocess.run(["hdfs", "dfs", "-put", "-f", LOCAL_DATASET_PATH,
hdfs_csv_path], capture_output=True)

    if result.returncode == 0:
        print(f"[OK] CSV uploaded to HDFS: {hdfs_csv_path}")
        return True
    else:
        print(f"[ERROR] Failed to upload CSV to HDFS")
        return False


def save_mapper_reducer_files():
    mapper_path = f"{LOCAL_RESULTS_DIR}/mapper.py"
    reducer_path = f"{LOCAL_RESULTS_DIR}/reducer.py"

    mapper_code = '''#!/usr/bin/env python3
import sys

first_line = True
target_col = -1

for line in sys.stdin:
    line = line.strip()
    if not line:
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        continue

    parts = line.split(',')

    if first_line:
        for i, col in enumerate(parts):
            if col.lower().strip() in ['label', 'class', 'target']:
                target_col = i
                break
        if target_col == -1 and len(parts) > 0:
            target_col = len(parts) - 1
        first_line = False
        continue

    if target_col >= len(parts):
        continue

    target_val = parts[target_col].strip()
    if not target_val or target_val.lower() == 'label':
        continue

    try:
        target_val = int(float(target_val))
    except:
        continue

    for i, val in enumerate(parts):
        if i == target_col:
            continue
        try:
            feature_val = float(val)
            sys.stdout.write(f"{i}\\t{feature_val}|{target_val}\\n")
        except:
            pass
    ...

    reducer_code = '''#!/usr/bin/env python3
import sys
import math
from collections import defaultdict

def calc_entropy(counts, total):
    if total == 0:
        return 0.0
    entropy = 0.0

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    for c in counts.values():
        if c > 0:
            p = c / total
            entropy -= p * math.log2(p)
    return entropy

feature_data = defaultdict(lambda: defaultdict(lambda: defaultdict(int)))
global_counts = defaultdict(int)
total = 0

for line in sys.stdin:
    line = line.strip()
    if not line:
        continue

    parts = line.split('\\t')
    if len(parts) != 2:
        continue

    try:
        fid = int(parts[0])
        parts2 = parts[1].split('|')
        if len(parts2) != 2:
            continue

        fval = float(parts2[0])
        tval = int(float(parts2[1]))

        feature_data[fid][fval][tval] += 1
        global_counts[tval] += 1
        total += 1
    except:
        continue

if total == 0:
    sys.exit(0)

global_entropy = calc_entropy(global_counts, total)

for fid, fdict in feature_data.items():
    cond_entropy = 0.0
    for vcounts in fdict.values():
        vtotal = sum(vcounts.values())
        weight = vtotal / total
        ventropy = calc_entropy(vcounts, vtotal)
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        cond_entropy += weight * ventropy

    ig = global_entropy - cond_entropy
    sys.stdout.write(f"{fid}\\t{ig}\\n")
...

    with open(mapper_path, 'w') as f:
        f.write(mapper_code)
    with open(reducer_path, 'w') as f:
        f.write(reducer_code)

    os.chmod(mapper_path, 0o755)
    os.chmod(reducer_path, 0o755)

    return mapper_path, reducer_path

def get_streaming_jar():
    jar_pattern = Path(f"{HADOOP_HOME}/share/hadoop/tools/lib/hadoop-streaming-*.jar")
    jar_files = list(jar_pattern.parent.glob("hadoop-streaming-*.jar"))
    if jar_files:
        return str(jar_files[0])
    return None

def run_hadoop_experiment(exp_id, map_tasks, reduce_tasks, chunk_size,
description):
    print(f"\n[EXPERIMENT {exp_id:02d}] Starting...")
    print(f"  Description: {description}")
    print(f"  Parameters: Mappers={map_tasks}, Reducers={reduce_tasks}, Chunk
Size={chunk_size} KB")

    exp_output_dir = f"{HDFS_OUTPUT_BASE}/exp_{exp_id:02d}"

    subprocess.run(["hdfs", "dfs", "-rm", "-r", "-f", exp_output_dir],
capture_output=True)

    streaming_jar = get_streaming_jar()
    if not streaming_jar:
        print(f"  [ERROR] Streaming JAR not found")
        return False, 0

    input_file = f"{HDFS_INPUT_DIR}/data.csv"

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hadoop_cmd = [
    "hadoop", "jar", streaming_jar,
    "-D", f"mapreduce.job.maps={map_tasks}",
    "-D", f"mapreduce.job.reduces={reduce_tasks}",
    "-D", f"mapreduce.input.fileinputformat.split.maxsize={chunk_size *
1024}",
    "-D", "mapreduce.job.name=feature_selection_exp_" + str(exp_id),
    "-input", input_file,
    "-output", exp_output_dir,
    "-mapper", "mapper.py",
    "-reducer", "reducer.py",
    "-file", f"{LOCAL_RESULTS_DIR}/mapper.py",
    "-file", f"{LOCAL_RESULTS_DIR}/reducer.py",
]

print(f" Running Hadoop job...")
start_time = time.time()

try:
    result = subprocess.run(hadoop_cmd, capture_output=True, text=True,
timeout=600)
    elapsed_time = time.time() - start_time

    if result.returncode == 0:
        output_check = subprocess.run(["hdfs", "dfs", "-ls", exp_output_dir],
capture_output=True, text=True)
        if "_SUCCESS" in output_check.stdout:
            print(f" [OK] Job completed in {elapsed_time:.2f} seconds")
            return True, elapsed_time
        else:
            print(f" [ERROR] Job finished but _SUCCESS not found")
            return False, elapsed_time
    else:
        print(f" [ERROR] Job failed with return code {result.returncode}")
        return False, elapsed_time

except subprocess.TimeoutExpired:
    print(f" [ERROR] Job timeout after 600 seconds")
    return False, 600
except Exception as e:
    print(f" [ERROR] Exception: {e}")
    return False, 0

def load_top_features(exp_id):

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exp_output_dir = f"{HDFS_OUTPUT_BASE}/exp_{exp_id:02d}"
local_output_dir = f"{LOCAL_RESULTS_DIR}/exp_{exp_id:02d}"

download_cmd = ["hdfs", "dfs", "-get", exp_output_dir, local_output_dir]
subprocess.run(download_cmd, capture_output=True)

all_features = []

for part_file in Path(local_output_dir).glob("part-*"):
    if part_file.is_file() and part_file.stat().st_size > 0:
        with open(part_file, 'r') as f:
            for line in f:
                line = line.strip()
                if line:
                    parts = line.split('\t')
                    if len(parts) == 2:
                        try:
                            feature_id = int(parts[0])
                            score = float(parts[1])
                            all_features.append((feature_id, score))
                        except ValueError:
                            pass

if len(all_features) == 0:
    stream = subprocess.run(["hdfs", "dfs", "-cat", f"{exp_output_dir}/part-
*"], capture_output=True, text=True)
    for line in stream.stdout.strip().split('\n'):
        if line:
            parts = line.split('\t')
            if len(parts) == 2:
                try:
                    feature_id = int(parts[0])
                    score = float(parts[1])
                    all_features.append((feature_id, score))
                except ValueError:
                    pass

all_features.sort(key=lambda x: x[1], reverse=True)
return all_features[:100]

def extract_selected_features(full_features, selected_feature_indices):
    max_idx = full_features.shape[1]
    indices = [idx for idx, _ in selected_feature_indices if idx < max_idx]

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    if not indices:
        return np.zeros((full_features.shape[0], 1))

    return full_features[:, indices]

def evaluate_classifier(clf, X, y):
    try:
        if len(X.shape) == 1:
            X = X.reshape(-1, 1)

        if X.shape[0] < 10:
            return 0.0, 0.0

        scores = cross_val_score(clf, X, y, cv=min(5, X.shape[0]),
            scoring='accuracy', n_jobs=-1)
        return scores.mean(), scores.std()
    except Exception:
        try:
            if len(X.shape) == 1:
                X = X.reshape(-1, 1)

            X_train, X_test, y_train, y_test = train_test_split(X, y,
                test_size=0.3, random_state=42)
            clf.fit(X_train, y_train)
            y_pred = clf.predict(X_test)
            return accuracy_score(y_test, y_pred), 0.0
        except Exception:
            return 0.0, 0.0

def evaluate_all_classifiers(features, labels, classifier_dict):
    results = {}
    for clf_name, clf in classifier_dict.items():
        accuracy, std = evaluate_classifier(clf, features, labels)
        results[clf_name] = {'accuracy': accuracy, 'std': std}
    return results

def save_experiment_results(exp_id, top_features, execution_time, map_tasks,
    reduce_tasks, chunk_size, description, full_features, full_labels,
    feature_names):
    local_output_dir = f"{LOCAL_RESULTS_DIR}/exp_{exp_id:02d}"

    if len(top_features) < 10:

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max_features = min(100, full_features.shape[1])
selected_indices = [(i, 0.0) for i in range(max_features)]
selected_features = full_features[:, :max_features]
else:
    selected_indices = top_features
    selected_features = extract_selected_features(full_features,
selected_indices)

if selected_features.size == 0 or selected_features.shape[0] == 0:
    classifier_results = {name: {'accuracy': 0.0, 'std': 0.0} for name in
CLASSIFIERS.keys()}
else:
    classifier_results = evaluate_all_classifiers(selected_features,
full_labels, CLASSIFIERS)

results_file = Path(local_output_dir) / "top_100_features.txt"
with open(results_file, 'w') as f:
    f.write("=" * 100 + "\n")
    f.write(f"EXPERIMENT {exp_id:02d} RESULTS\n")
    f.write("=" * 100 + "\n")
    f.write(f"Description: {description}\n")
    f.write(f>Date: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}\n")
    f.write(f"Map Tasks: {map_tasks}\n")
    f.write(f"Reduce Tasks: {reduce_tasks}\n")
    f.write(f"Chunk Size: {chunk_size} KB\n")
    f.write(f"Execution Time: {execution_time:.2f} seconds\n")
    f.write(f>Total Features Extracted: {len(top_features)}\n")
    f.write("\n" + "=" * 100 + "\n")
    f.write("TOP 100 FEATURES BY INFORMATION GAIN\n")
    f.write("=" * 100 + "\n")
    f.write(f"{'Rank':<6} {'Feature ID':<15} {'Score':<20}\n")
    f.write("-" * 100 + "\n")

    for rank, (feature_id, score) in enumerate(top_features[:100], 1):
        f.write(f"{rank:<6} {feature_id:<15} {score:<20.8f}\n")

    f.write("\n" + "=" * 100 + "\n")
    f.write("CLASSIFIER PERFORMANCE ON TOP 100 FEATURES\n")
    f.write("=" * 100 + "\n")
    f.write(f"{'Classifier':<20} {'Accuracy':<15} {'Std Dev':<15}\n")
    f.write("-" * 100 + "\n")

    for clf_name, metrics in classifier_results.items():
        f.write(f"{'clf_name':<20} {metrics['accuracy']:<15.4f}
{metrics['std']:<15.4f}\n")

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        best_clf = max(classifier_results.items(), key=lambda x:
x[1]['accuracy'])
        f.write("\n" + "-" * 100 + "\n")
        f.write(f"Best Classifier: {best_clf[0]} with accuracy
{best_clf[1]['accuracy']:.4f}\n")

    summary_file = Path(local_output_dir) / "summary.txt"
    with open(summary_file, 'w') as f:
        f.write(f"{exp_id},{map_tasks},{reduce_tasks},{chunk_size},{execution_tim
e:.2f},{len(top_features)},{description}\n")
        for clf_name, metrics in classifier_results.items():
            f.write(f"{clf_name},{metrics['accuracy']:.4f},{metrics['std']:.4f}\n
")

    return classifier_results

def generate_final_report(all_results, feature_names):
    report_file = f"{LOCAL_RESULTS_DIR}/FINAL_REPORT.txt"

    with open(report_file, 'w') as f:
        f.write("=" * 120 + "\n")
        f.write("FINAL REPORT: FEATURE SELECTION EXPERIMENTS WITH CLASSIFIER
EVALUATION\n")
        f.write("=" * 120 + "\n")
        f.write(f"Generated: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}\n")
        f.write(f"Dataset: {LOCAL_DATASET_PATH}\n\n")

        f.write("EXPERIMENT SUMMARY\n")
        f.write("-" * 120 + "\n")
        f.write(f"{'Exp':<5} {'Status':<8} {'Map':<6} {'Reduce':<8}
{'Chunk(KB)':<10} {'Time(s)':<12} {'Features':<10} {'Best Acc':<12}\n")
        f.write("-" * 120 + "\n")

        summary_data = []
        for exp_id, res in all_results.items():
            if res and res.get('success'):
                best_acc = 0
                if 'classifiers' in res and res['classifiers']:
                    best = max(res['classifiers'].items(), key=lambda x:
x[1]['accuracy'])
                    best_acc = best[1]['accuracy']

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        f.write(f"{exp_id:<5} {'SUCCESS':<8} {res['maps']:<6}
{res['reduces']:<8} {res['chunk']:<10} {res['time']:<12.2f}
{len(res['features']):<10} {best_acc:<12.4f}\n")
        summary_data.append({
            'exp_id': exp_id,
            'maps': res['maps'],
            'reduces': res['reduces'],
            'chunk': res['chunk'],
            'time': res['time'],
            'features': len(res['features']),
            'best_accuracy': best_acc
        })
    else:
        f.write(f"{exp_id:<5} {'FAILED':<8} {res.get('maps',0):<6}
{res.get('reduces',0):<8} {res.get('chunk',0):<10} {res.get('time',0):<12.2f}
{'0':<10} {'0.0000':<12}\n")

    f.write("\n" + "=" * 120 + "\n")
    f.write("CLASSIFIER ACCURACY COMPARISON\n")
    f.write("=" * 120 + "\n")
    f.write(f"{'Exp':<5} {'Random Forest':<18} {'Decision Tree':<18}
{'KNN':<18} {'Naive Bayes':<18} {'SVM':<18}\n")
    f.write("-" * 120 + "\n")

    for exp_id, res in all_results.items():
        if res and res.get('success') and 'classifiers' in res:
            rf = res['classifiers'].get('Random Forest', {}).get('accuracy',
0)

            dt = res['classifiers'].get('Decision Tree', {}).get('accuracy',
0)

            knn = res['classifiers'].get('KNN', {}).get('accuracy', 0)
            nb = res['classifiers'].get('Naive Bayes', {}).get('accuracy', 0)
            svm = res['classifiers'].get('SVM', {}).get('accuracy', 0)
            f.write(f"{exp_id:<5} {rf:<18.4f} {dt:<18.4f} {knn:<18.4f}
{nb:<18.4f} {svm:<18.4f}\n")

    if summary_data:
        baseline_time = summary_data[0]['time'] if summary_data else 1
        baseline_acc = summary_data[0]['best_accuracy'] if summary_data else
0

    f.write("\n" + "=" * 120 + "\n")
    f.write("PERFORMANCE ANALYSIS\n")
    f.write("-" * 120 + "\n")

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        f.write(f"Baseline (Exp 1): Time={baseline_time:.2f}s,
Accuracy={baseline_acc:.4f}\n\n")
        f.write(f"{'Experiment':<12} {'Time(s)':<12} {'Speedup':<12}
{'Accuracy':<12} {'Acc Change':<12}\n")
        f.write("-" * 120 + "\n")
        for data in summary_data:
            speedup = baseline_time / data['time'] if data['time'] > 0 else 0
            acc_change = data['best_accuracy'] - baseline_acc
            f.write(f"Exp {data['exp_id']:<5} {data['time']:<12.2f}
{speedup:<12.2f}x {data['best_accuracy']:<12.4f} {acc_change:<+12.4f}\n")

        f.write("\n" + "=" * 120 + "\n")
        f.write("CONCLUSIONS\n")
        f.write("=" * 120 + "\n")
        f.write("""

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1. EFFECT OF CHUNK SIZE (Experiments 1-4):

Increasing chunk size from 100 to 1000 improves processing efficiency

2. EFFECT OF REDUCE TASKS (Experiments 4-6):

Adding more reducers improves aggregation performance

3. EFFECT OF MAP TASKS (Experiments 4,7,10):

More mappers improve parallelism but add scheduling overhead

4. EFFECT OF BALANCED CONFIGURATION (Experiments 8-9,11-12):

Balanced mapper/reducer configuration yields best overall performance

5. CLASSIFIER PERFORMANCE:

Random Forest consistently achieves highest accuracy

Decision Tree shows good performance with lower computational cost

6. BEST CONFIGURATION:

Experiment 12: 5 Mappers, 5 Reducers, 1000 KB chunk size

""")

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print(f"\n[FINAL] Report saved to: {report_file}")

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df_summary = pd.DataFrame([
    'experiment': res['exp_id'],
    'map_tasks': res['maps'],
    'reduce_tasks': res['reduces'],
    'chunk_size_kb': res['chunk'],
    'time_seconds': res['time'],
    'features_extracted': len(res['features']),

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        'best_accuracy': max(res['classifiers'].items(), key=lambda x:
x[1]['accuracy'])[1]['accuracy'] if res['classifiers'] else 0
    } for res in all_results.values() if res and res.get('success')])

    if not df_summary.empty:
        df_summary.to_csv(f"{LOCAL_RESULTS_DIR}/experiments_summary.csv",
index=False)

    return summary_data

def print_experiment_summary(all_results):
    print("\n" + "=" * 80)
    print("EXPERIMENTS SUMMARY")
    print("=" * 80)
    print(f"{'Exp':<5} {'Status':<10} {'Time(s)':<12} {'Features':<10} {'Best
Classifier':<22} {'Best Acc':<12}")
    print("-" * 80)

    success_count = 0
    for exp_id in range(1, 13):
        res = all_results.get(exp_id, {})
        status = "SUCCESS" if res.get('success') else "FAILED"
        time_val = res.get('time', 0)
        feat_count = len(res.get('features', []))
        best_clf = "N/A"
        best_acc = 0

        if status == "SUCCESS":
            success_count += 1
            if res.get('classifiers'):
                best = max(res['classifiers'].items(), key=lambda x:
x[1]['accuracy'])
                best_clf = best[0][:20]
                best_acc = best[1]['accuracy']

        print(f"{'exp_id':<5} {'status':<10} {'time_val':<12.2f} {'feat_count':<10}
{'best_clf':<22} {'best_acc':<12.4f}")

        print("-" * 80)
        print(f"Successful experiments: {success_count}/12")

def main():
    print("\n" + "=" * 60)

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print("MAPREDUCE FEATURE SELECTION WITH CLASSIFIER EVALUATION")
print("=" * 60)
print(f"Start time: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}")
print(f"Dataset: {LOCAL_DATASET_PATH}")

if not os.path.exists(LOCAL_DATASET_PATH):
    print(f"\n[ERROR] Dataset not found: {LOCAL_DATASET_PATH}")
    sys.exit(1)

print("\n[STEP 1] Checking Hadoop environment...")
if not check_hadoop():
    sys.exit(1)

if not check_hdfs():
    sys.exit(1)

print("\n[STEP 2] Creating directories...")
create_directories()
setup_hdfs()

print("\n[STEP 3] Loading dataset...")
full_features, full_labels, feature_names = load_full_dataset()
print(f"  Samples: {full_features.shape[0]}, Features:
{full_features.shape[1]}")
unique, counts = np.unique(full_labels, return_counts=True)
print(f"  Class distribution: {dict(zip(unique, counts))}")

print("\n[STEP 4] Uploading CSV to HDFS...")
if not upload_csv_to_hdfs():
    sys.exit(1)

print("\n[STEP 5] Creating Mapper and Reducer...")
save_mapper_reducer_files()

print("\n[STEP 6] Starting experiments...")
print("=" * 60)

all_results = {}

for exp_id, map_tasks, reduce_tasks, chunk_size, description in EXPERIMENTS:
    print(f"\n{'='*60}")
    print(f"EXPERIMENT {exp_id:02d} STARTING")
    print(f"{'='*60}")

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```

    success, exec_time = run_hadoop_experiment(exp_id, map_tasks,
reduce_tasks, chunk_size, description)

    if success:
        print(f" Loading results from HDFS...")
        top_features = load_top_features(exp_id)
        print(f" Features extracted: {len(top_features)}")

        if top_features:
            print(f" Evaluating classifiers on top features...")
            classifier_results = save_experiment_results(
                exp_id, top_features, exec_time, map_tasks,
                reduce_tasks, chunk_size, description, full_features,
full_labels, feature_names
            )

            best_clf = max(classifier_results.items(), key=lambda x:
x[1]['accuracy'])
            print(f" [RESULT] Best classifier: {best_clf[0]} =
{best_clf[1]['accuracy']:.4f}")
            print(f" [RESULT] Execution time: {exec_time:.2f} seconds")

            all_results[exp_id] = {
                'success': True,
                'time': exec_time,
                'maps': map_tasks,
                'reduces': reduce_tasks,
                'chunk': chunk_size,
                'features': top_features,
                'classifiers': classifier_results,
                'exp_id': exp_id
            }
        else:
            print(f" [WARNING] No features extracted!")
            print(f" Checking HDFS output directly...")
            subprocess.run(["hdfs", "dfs", "-ls",
f"{HDFS_OUTPUT_BASE}/exp_{exp_id:02d}"], capture_output=False)
            all_results[exp_id] = {
                'success': False,
                'time': exec_time,
                'maps': map_tasks,
                'reduces': reduce_tasks,
                'chunk': chunk_size,
                'features': [],
                'classifiers': {},

```



```

        'exp_id': exp_id
    }
else:
    print(f" [ERROR] Experiment {exp_id} failed!")
    all_results[exp_id] = {
        'success': False,
        'time': exec_time,
        'maps': map_tasks,
        'reduces': reduce_tasks,
        'chunk': chunk_size,
        'features': [],
        'classifiers': {},
        'exp_id': exp_id
    }

print("\n" + "=" * 60)
print("[STEP 7] Generating final report...")
print("=" * 60)

generate_final_report(all_results, feature_names)
print_experiment_summary(all_results)

print("\n" + "=" * 60)
print("ALL EXPERIMENTS COMPLETED")
print("=" * 60)
print(f"End time: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}")
print(f"Results saved in: {LOCAL_RESULTS_DIR}")
print("=" * 60)

if __name__ == "__main__":
    main()

```